

Business Task

The objective of this analysis is to identify behavioral differences between annual members and casual riders using Cyclistic's historical trip data. These insights will help Cyclistic's marketing team develop strategies to convert casual riders into annual members.

Data Sources

Data source:

Divvy trip data (Chicago's bike-share system), publicly available at divvy-tripdata.s3.amazonaws.com, licensed by Motivate International Inc.

The original dataset contained approximately 5.9 million trips collected between July 2015 and June 2016. After data cleaning, 5,770,091 trips remained for analysis.

ROCCC?

Reliable:

Large sample size (5,770,091 cleaned trips), providing sufficient observations for reliable analysis.

Original:

Official Divvy source, not third-party aggregated.

Comprehensive:

Contains all fields needed to answer the business question.

Current:

Most recent 12 months available.

Cited:

Publicly documented source with clear licensing.

Limitation:

Personally identifiable information is excluded per Divvy's privacy policy, so casual riders cannot be linked to Cyclistic's service area or verified as repeat single_pass buyers.

Data Cleaning:

Removed 29 trips with negative duration.

Removed 162,246 trips shorter than 60 seconds which likely represented false starts, or improperly docked bicycles.

Removed 12 trips longer than 24 hours, which were considered unrealistic ride durations.

Final cleaned dataset: 5,770,091 trips.

Key Findings

Ride Length:

Casual riders average 22.6 minutes per ride, compared to 12.7 minutes for members, suggesting that casual riders primarily use bikes for leisure, while members rely on them for shorter, routine trips.

Day of Week:

Members recorded their highest number of trips during weekdays, especially Tuesday through Thursday, indicating commuting behavior. Casual riders showed increased activity during weekends, particularly on Saturdays, suggesting recreational usage.

Seasonality:

Both groups peak in the summer months, but casual usage drops off almost entirely in winter, while members maintain steadier (if lower) usage level year-round.

Bike Type:

Members account for a higher total number of rides on both classic and electric bikes. Electric bikes were the preferred bike type for both rider groups.

Recommendations

Weekend-focused membership campaign:

Because casual riders take their longest and most frequent rides during weekends, Cyclistic should launch weekend membership promotions targeting leisure cyclists.

Seasonal membership options:

Introduce a summer half-year membership (e.g., April-September) and/or a flexible monthly membership aimed at casual riders. Since casual usage is highly seasonal (peaking in summer and dropping to almost nothing in winter), this gives them an option that better matches their actual pattern than a standard annual membership, lowering the barrier to converting from casual to paying member.

Electric bike as a membership perk:

Use free or discounted access to electric bikes as a concrete incentive when marketing to casual riders, since both groups already prefer this bike type.

Target frequent casual riders:

Offer targeted discounts to frequent casual riders who regularly use electric bikes during summer weekends, encouraging them to convert to annual membership.

Conclusion

This analysis demonstrates a clear behavioral difference between annual members and casual riders. Members primarily use Cyclistic bikes for routine transportation, while casual riders ride longer and show stronger weekend and seasonal usage patterns. These insights provide opportunities for targeted marketing campaigns designed to convert casual riders into annual members and support Cyclistic's long-term growth strategy.

Interactive Dashboard

View the interactive Tableau dashboard here:

https://public.tableau.com/app/profile/stian.landell/viz/CyclisticBikeShareAnalysisDashboard_17845442035820/Cyclistic_Dashboard?publish=yes